

Kishan Prajapati

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Education

Northeastern University, Khoury College of Computer Sciences

January 2026 – May 2028

Master of Science, Computer Science, Boston, MA

P P Savani University

October 2021 – May 2025

B.Tech Information Technology, First Class with Distinction. Top grade in Algorithms, Data Structures, Discrete Math

Experience

Graduate Teaching Assistant

May 2026 – Present

Khoury College of Computer Sciences, Boston, MA

- Selected for **three consecutive appointments** by three faculty across data science, algorithms, and discrete structures.
- Lead a small-group tutoring cohort, communicating analytical reasoning to students at very different levels.
- Introduced a cascade-error grading convention adopted in collaboration with the TA team across **200+ students**.
- Review student project proposals, flagging an intractable TSP formulation before it cost a student a semester.

Software Developer Intern

December 2024 – March 2025

Technonite, Surat, India

- Built a reusable React component library across **3 client projects**, cutting feature delivery time ~25%.
- Shipped a **real-time** Socket.io logistics dashboard to production on release cycles measured in days.

Undergraduate Researcher

November 2023 – March 2025

P P Savani University, Surat, India

- Built the ML pipeline for a **Springer-published** study at **88.10% accuracy**, beating the prior benchmark by 8 points.
- Ran Naive Bayes, SVM and logistic regression over TF-IDF, LIWC and sentiment features on 20K labelled records.
- Constructed a **49,000-row** dataset across **490+ event categories** and shipped a live Hugging Face demo.

Competitive Programming Head

August 2022 – September 2023

Google Developer Student Club, Surat, India

- Led the competitive programming vertical: **15+ contests** on Codeforces and LeetCode, **1,000+ participants**.
- Built and taught a weekly algorithms and data structures curriculum for **50+ students**.

Projects

Gatekeepn't: Reference-Bias Audit of a Standard Evaluation Metric

Python, PyTorch, QLoRA, Hugging Face, TRL, NVIDIA A100 · First author, under ACL Rolling Review

- Showed a metric used across the field **rewards imitation, not skill: -0.84 Spearman** over a 28-cell design.
- Trained **Llama 3.1 8B with QLoRA** on 239K examples, plus BioBART-v2 and FLAN-T5, on one A100.
- Quantified the cost: acting on the metric scores **15 points below the do-nothing baseline**.
- Isolated the cause in the metric, not the training, by testing cases where training could not explain it.

Sieve: Hybrid Retrieval and Deduplication over 183K Records

Python, FastAPI, PostgreSQL, pgvector, ONNX Runtime, React, Docker

- Built the whole system solo: ingestion, Postgres and pgvector retrieval behind a REST API, and a React front end.
- Turned a **quadratic** self-join into GROUP BY aggregation with **union-find**, cutting 11 minutes to seconds.
- Found an **1,875x** performance regression, 1,140 ms to 0.608 ms, from a missing index, by reading the query plan.
- Backed it with **118 unit and integration tests** on GitHub Actions CI and 14,135 duplicate records reconciled.
- Built a benchmark harness refusing unsupportable statistics, which corrected six of my own earlier measurements.

Technical Skills

Languages: Java, C++, Python, TypeScript, JavaScript, SQL, Bash

Core CS: Data structures, algorithms, object-oriented programming, software design, unit and integration testing

Systems: FastAPI, REST APIs, PostgreSQL, pgvector, Docker, Linux, Git, GitHub Actions, pytest, EXPLAIN ANALYZE

ML and Tooling: PyTorch, scikit-learn, Hugging Face, Pandas, NumPy, AI coding assistants (Claude Code)

Publications

J. Patil, K. Prajapati, et al. Transforming AI for Depression Detection: A Review. *CML 2025*, Springer LNNS, Scopus.

J. Patil, V. Patil, K. Prajapati, et al. Enhanced Depression Detection on Social Media. *ICICC 2024*, Springer LNNS.

M. Kashyap, R. Chauhan, D. Patel, K. Prajapati. Green Web 3.0, Decentralized AI, Edge Intelligence. *ITM Web Conf. 2024*.